

# Dhanush Kumar K

+91 8220641028 · dhanushkumark.tech@gmail.com · LinkedIn · GitHub · Coimbatore, India

## Professional Summary

Final-year B.Tech student in Artificial Intelligence & Data Science with hands-on experience designing scalable software systems, RESTful APIs, and intelligent applications through internships, hackathons, and independent projects. Proficient in Java, Python, Spring Boot, React, SQL, and modern software engineering practices. Winner of DevHack 2026 and solved 200+ LeetCode problems, with a strong interest in backend engineering, distributed systems, and high-performance applications.

## Technical Skills

|                        |                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------|
| Languages              | Python, Java, C, C++, SQL                                                                         |
| Frameworks & Libraries | Spring Boot 3, Flask, FastAPI, React, Tailwind CSS, Axios                                         |
| Databases              | MySQL, SQLite                                                                                     |
| Data & ML              | Pandas, NumPy, Matplotlib · Scikit-learn, XGBoost · Power BI, DAX                                 |
| Developer Tools        | Docker, Git, GitHub, Postman                                                                      |
| Security               | Spring Security, JWT                                                                              |
| Concepts               | OOP, DSA, RESTful APIs, Authentication, MVC, Database Design, Layered Architecture, System Design |

## Experience

**Data Analytics Intern** May 2025 – Jun 2025  
*Accent Techno Soft* Coimbatore, India

- Built 3 Power BI dashboards using Power Query and DAX on 10,000+ row retail datasets, automating KPI reporting and eliminating manual weekly Excel workflows for business stakeholders.
- Developed an ETL pipeline using Python and Pandas to clean and standardise six months of raw sales data across four product categories, improving downstream data reliability and consistency.

## Projects

**CookieStore Management System** | *Java 17, Spring Boot 3, MySQL, React, JWT* GitHub

- Built a production-ready retail platform with 8-role RBAC and JWT authentication; designed 40+ stateless REST APIs using layered architecture (Controller, Service, Repository, DTO) covering order lifecycle, inventory, and batch scheduling.
- Implemented global exception handling, enum-driven workflow states, Jest frontend testing, and responsive dark/light theme support.

**MASTER** | *Python, Flask, React, Pandas, SQLite, Ollama* GitHub

- Designed and developed a full-stack multi-agent simulation with 6 autonomous trading agents operating on live financial market data (Yahoo Finance API), a Flask REST backend with 9 API endpoints, and a React frontend with TradingView-style candlestick charts.
- Engineered a compliance engine with 5 compliance rules and circuit breakers blocking 37+ violations per run; integrated a local LLM agent via Ollama for zero-cloud-cost decisions. Won 1st place among 160+ teams at DevHack 2026.

**NameFlux** | *C++, Python, Streamlit, Bloom Filter, LRU Cache* GitHub

- Designed a persistent hybrid architecture with a C++ core engine combining a custom LRU Cache and Bloom Filter, paired with a Python/Streamlit monitoring frontend, simulating real-time username lookup across 1M+ records.
- Implemented a three-layer query resolution path achieving benchmark-tested cache lookup latency below 15 µs; built a live monitoring dashboard tracking cache hit rate and false-positive count in real time.

**FIFA World Cup 2026 Predictor** | *Python, XGBoost, Scikit-learn, Pandas, Monte Carlo* GitHub

- Built an end-to-end ML pipeline trained on 49,000+ historical matches using Elo ratings, rolling form, and xG features; ran 10,000 Monte Carlo simulations to forecast champion and group-qualification probabilities.
- Improved prediction accuracy by 22 percentage points over a random baseline (55% vs. 33%) using XGBoost and historical match features; deployed a self-contained HTML dashboard with automatic hourly live-result updates.

## Education

**B.Tech, Artificial Intelligence & Data Science** 2023 – 2027 (Expected)  
*Sri Krishna College of Technology* Coimbatore, India

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Organisation, Operating Systems, Computer Networks, Object-Oriented Programming

## Achievements

**1st Place — DevHack Hackathon 2026** | 160+ teams | Sri Krishna College of Technology

**LeetCode** | 200+ problems solved | Arrays, Strings, Trees, Graphs, Dynamic Programming, SQL

**Kaggle — Titanic Survival Prediction** | Leaderboard score: **0.86555** | Gradient Boosting pipeline

## Certifications

**Cisco** | Python Essentials 1 & 2

**AWS Educate** | Machine Learning Foundations

**MongoDB** | Schema Design